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Spam Message Classification Using Machine Learning

Abstract

Spam messages are one of the most common problems in digital communication systems. Spam messages contain advertisements, fraudulent links, fake offers, or malicious content that may affect users and organizations. This project focuses on developing a Machine Learning based Spam Message Classification System that can classify messages as either Spam or Ham (non-spam). The system uses text preprocessing techniques, TF-IDF vectorization, and Logistic Regression algorithm for accurate classification. A Flask web application is also developed to provide a user-friendly interface where users can enter messages and instantly receive predictions.

Introduction

In today's digital world, millions of text messages are exchanged every day through emails, SMS, and social media platforms. Among these messages, many are spam messages that contain fake advertisements, phishing links, scams, or harmful information. Detecting such spam messages manually is difficult and time-consuming. Therefore, Machine Learning techniques are widely used to automate spam message detection.

Spam message classification is a Natural Language Processing (NLP) application where textual data is analyzed and categorized. The purpose of this project is to build a web-based application capable of identifying whether a message is spam or ham using Machine Learning algorithms.

The developed system preprocesses the text, converts text into numerical features using TF-IDF Vectorization, trains a Logistic Regression model, and predicts whether the input message is spam or ham.

Objectives

The main objectives of this project are:

- To develop a machine learning model for spam message classification.
- To preprocess and clean textual data.
- To convert text into numerical vectors using TF-IDF.

- To train the model using Logistic Regression algorithm.
- To build a web application using Flask.
- To provide real-time spam or ham prediction.
- To improve communication security and reduce spam-related risks.

Problem Statement

Spam messages are increasing rapidly and can cause problems such as fraud, phishing attacks, malware distribution, and unnecessary advertisements. Traditional filtering methods are not efficient enough to handle modern spam messages. Therefore, there is a need for an intelligent system that can automatically classify spam and ham messages accurately using Machine Learning techniques.

Dataset Description

The dataset used for this project is the SMS Spam Collection Dataset obtained from Kaggle.

Dataset Link:
https://www.kaggle.com/datasets/uciml/sms-spam-collection-dataset

The dataset contains two columns:

1. Label

- Indicates whether the message is spam or ham.
- Spam = unwanted message
- Ham = genuine message

2. Text

- Contains the actual message content.

Example Dataset:

Label	Text
ham	Hey, are we meeting tomorrow?
spam	Congratulations! You won a free iPhone.

The dataset contains thousands of real-world messages which help improve the model performance.

Technologies Used

The following technologies and tools were used in this project:

Technology	Purpose
Python	Programming Language
Flask	Web Framework
Pandas	Data Handling
Scikit-learn	Machine Learning
NLTK	Text Preprocessing
HTML/CSS	Frontend Design
TF-IDF	Feature Extraction
Logistic Regression	Classification Algorithm

System Architecture

The overall workflow of the project is:

1. User enters a message.
2. Message is sent to Flask backend.
3. Text preprocessing is applied.
4. TF-IDF converts text into vectors.
5. Logistic Regression model predicts the category.
6. Result is displayed as Spam or Ham.

Text Preprocessing

Text preprocessing is an important step in Natural Language Processing because raw text contains unnecessary symbols and stopwords.

The preprocessing steps used in this project are:

1. Lowercase Conversion

All text is converted into lowercase to maintain consistency.

Example:

“FREE OFFER” → “free offer”

2. Removing Punctuation

Special characters and punctuation marks are removed.

Example:

“Win money!!!” → “Win money”

3. Stopword Removal

Common words such as “the”, “is”, “are” are removed because they do not contribute significantly to classification.

Feature Extraction Using TF-IDF

Machine Learning models cannot directly understand text data. Therefore, text messages are converted into numerical vectors using TF-IDF Vectorization.

TF-IDF stands for:

- TF = Term Frequency
- IDF = Inverse Document Frequency

TF-IDF gives more importance to important words and reduces the importance of common words.

Example:

Words like “free”, “winner”, “prize” are highly important in spam detection.

Advantages of TF-IDF:

- Efficient text representation
- Reduces noise
- Improves classification accuracy

Machine Learning Algorithm

Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for classification problems.

In this project, Logistic Regression predicts whether the message belongs to:

- Spam

- Ham

Why Logistic Regression?

- Simple and efficient
- Works well for text classification
- High prediction accuracy
- Fast training and prediction

Working Principle

The algorithm calculates probabilities and classifies the message based on learned patterns from training data.

Model Training

The model training process includes:

1. Loading the dataset
2. Cleaning the text data
3. Converting labels into numerical format
4. Applying TF-IDF vectorization
5. Training Logistic Regression model
6. Saving trained model using Pickle

The trained model is saved as:

- model.pkl
- vectorizer.pkl

Web Application Development

A Flask web application is developed to provide a user-friendly interface.

Features of the web application:

- Attractive UI Design
- Real-time message classification
- Spam/Ham prediction
- Responsive interface

Frontend Technologies:

- HTML
- CSS

Backend Technology:

- Flask

Project Folder Structure

```
spam-classifier/
│
├── app.py
├── train_model.py
├── spam.csv
├── model.pkl
├── vectorizer.pkl
├── README.md
│
├── templates/
│   └── index.html
│
└── static/
└── style.css
```

Advantages of the System

- Fast message classification
- Reduces spam-related risks
- User-friendly interface
- Real-time prediction
- Easy deployment
- Accurate spam detection

Limitations

- Small datasets reduce accuracy.
- Cannot detect highly advanced spam patterns.

- Requires retraining for updated spam trends.

Future Enhancements

The project can be enhanced in future using:

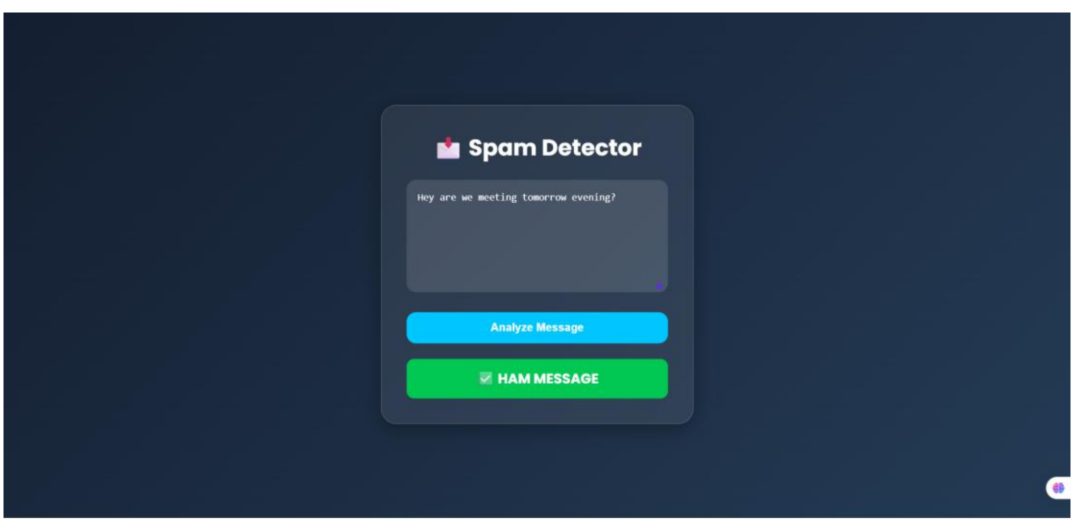
- Deep Learning algorithms
- LSTM and RNN models
- Email spam detection
- Multi-language support
- Cloud deployment
- Mobile application integration

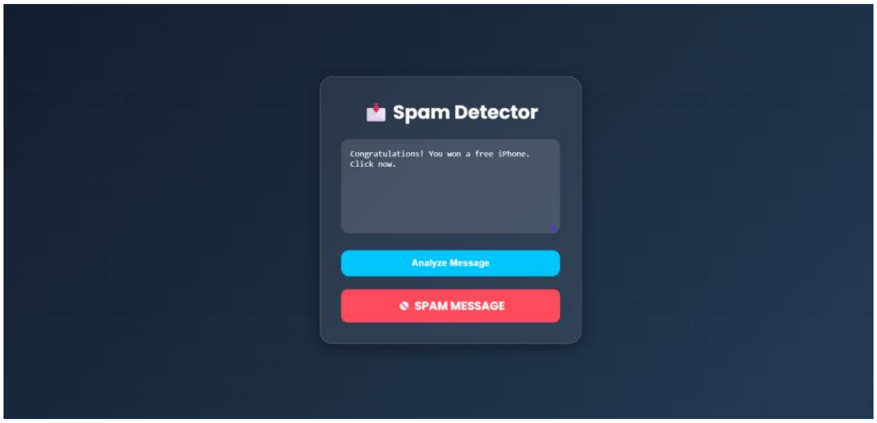
Output Screenshots

The following screenshots can be added:

1. Home Page
2. Spam Message Prediction
3. Ham Message Prediction
4. Model Training Output

Output





Conclusion

This project successfully developed a Spam Message Classification System using Machine Learning techniques. The system preprocesses text messages, converts them into numerical vectors using TF-IDF, and classifies them using Logistic Regression algorithm. A Flask-based web application provides real-time predictions through a simple and attractive user interface.

The developed model can effectively identify spam and ham messages, helping users avoid fraudulent or unwanted messages. This project demonstrates the practical implementation of Machine Learning and Natural Language Processing in real-world applications.
